A Novel Multi-Step Short-Term Power Forecasting Model for Electric Power Systems Based on Deep-Learning

Qiansheng Fang, Xinlong Liang, Shanshou Li, Jian Zhang, Xuelei Yang, and Zhangquanyi Guo

***Abstract*—Load forecasting and renewable energy forecasting are fundamental for the optimization and scheduling of new power systems, which play a crucial role in ensuring the safe, stable, and economical operation of the system. This paper proposes a short-term forecasting model based on Transformer-BiLSTM-Patch to meet the demand for short-term forecasting of load and renewable energy. The model designs Patch Embedding to improve attention calculation from traditional point-to-point interactions to segment-to-segment interactions, which is capable of capturing local trends. The model eliminates the mask mechanism during the decoding stage to fully utilize information from the entire time series, that is particularly effective in addressing scenarios specifically where auxiliary variables for future time steps are known (such as weather forecast data). Furthermore, BiLSTM is integrated into the Decoder to further enhance the model's capability in mining temporal features. Finally, comparative tests conducted on multiple datasets demonstrate that the proposed model significantly improves the performance of Transformer models in time series forecasting.**

***Index Terms*—Renewable energy forecasting, load forecasting, transformer, multi-step prediction.**

I. Introduction

In RECENT years, the global energy structure is undergoing a profound transformation. Renewable energies such as solar and wind have been widely applied. The new power system is formed on the basis of the traditional power system through technological innovation and institutional mechanism reform. It primarily consists of a high proportion of renewable energy to meet the electricity consumption needs of users. This system is characterized by clean and low-carbon, safe and efficient, and intelligently regulated [1].

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As renewable energy sources such as wind and solar power are increasingly utilized on a large scale, the inherent unpredictability, variability, and peak-shaving nature of renewable energy generation present substantial challenges to the stability and security of power systems [2]. In order to ensure the safe, reliable, efficient, and stable operation of new power systems, it is crucial to explore methods for short-term forecasting of renewable energy and load demand, which holds significant scientific and engineering value [3].

Power forecasting can be categorized into three types based on the duration of the forecasting time scale: long-term, short-term, and ultra-short-term forecasting. Long-term forecasting is applied in planning the scale of energy storage facilities, designing layouts for renewable energy generation systems, and formulating strategic plans for their construction and maintenance [4]. Short-term forecasting is utilized in daily scheduling and short-term energy storage planning [5], while ultra-short-term forecasting is employed for safety monitoring and emergency management [6]. Among these, short-term forecasting typically requires advanced forecasting models and algorithms, utilizing data inputs including electricity price information, historical generation statistics and Numerical Weather Prediction (NWP) to predict renewable energy output and load demand at 15-minute intervals (or finer) throughout the day. This provides essential decision support for grid operations, facilitating accurate energy management in new power systems. However, compared to the other two categories, short-term forecasting presents heightened challenges owing to the complexity of multi-step prediction [7], [8], making the enhancement of accuracy in short-term load forecasting and wind and solar power generation forecasting a critical focus in research and development.

In the field of power forecasting, traditional methods such as multivariate linear regression, Autoregressive Integrated Moving Average (ARIMA), Holt-Winters (HW), Markov Chains, support vector machines, radial basis function neural networks [9], [10], and others exhibit significant limitations in handling complex data and nonlinear relationships. Conversely, deep learning techniques, anchored in data-driven paradigms, exhibit significant advantages in revealing data correlations and enhancing model generalization capabilities, which have aroused extensive attention in the domain of power forecasting [11]. Long Short-Term Memory (LSTM) networks, as a type of nonlinear recurrent network, offer advantages over traditional

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neural networks in terms of temporal correlations and long-term memory capabilities. In recent years, LSTM has been widely applied and researched in the fields of time series analysis and forecasting. Chai et al. [12] enhanced the capability of LSTM networks in handling temporal correlations by introducing time-dependent learning weights, successfully applying this approach to the prediction of photovoltaic power generation. Jaihuni et al. [13] compared the performance of LSTM, Bidirectional Long Short-Term Memory (BiLSTM), and Bidirectional Gated Recurrent Unit (BiGRU) models with several commonly used models for predicting urban photovoltaic output. The results indicated that the BiLSTM model achieved the best performance. Liu et al. [14] integrated Convolutional Neural Networks (CNNs), BiLSTM, and attention mechanisms for ultra-short-term power load forecasting. Although the LSTM model has shown good performance in the field of ultra-short-term power forecasting, its step-by-step decoding prediction structure causes the model to lose feature information when dealing with long input sequences. Additionally, during long sequence time-series forecasting, the output of the model depends on the previous time step's predictions, leading to error accumulation. This phenomenon of error accumulation may affect prediction accuracy, resulting in decreased forecasting precision.

In 2017, Google introduced the Transformer neural network architecture [15], which is entirely based on the Self-Attention mechanism and has achieved significant progress in the fields of natural language processing, image recognition, and time series analysis. The unique Encoder-Decoder structure enables parallel computation and possesses strong capabilities in capturing long-range dependencies. The widespread adoption of the Transformer model has led to refinements and enhancements to its fundamental architecture, resulting in the emergence of multiple derivative models such as DETR, Informer, PatchTST [16], [17], [18].

Meanwhile, Transformers and their structural variants have also attracted considerable attention from researchers in photovoltaic generation forecasting, wind power generation forecasting, and load forecasting. Chen et al. [19] have employed improved Transformer models for load forecasting, The results showed that, compared with traditional technologies, not only has there been breakthrough progress in prediction accuracy, but also outstanding performance in terms of reasoning speed. Furthermore, Li et al. [20] enhanced the performance of load forecasting by employing the Particle Swarm Optimization (PSO) algorithm to fine-tune the hyperparameters of the Informer model. Simultaneously, Tian et al. [21] integrated CNN with Transformer technology to develop a day-ahead load interval prediction model, aiming to forecast the fluctuation range of load power for 48 time points within the upcoming day. In addition, when addressing more complex prediction tasks such as joint forecasting of multi-energy power, prediction of net load power intervals, probability forecasting of wind power, and transfer learning, some researchers have also incorporated Transformer as a key component in their model designs [22], [23], [24]. This highlights the powerful capability and flexibility of Transformers in handling different types of energy data and their complex relationships.

Although the Transformer model has brought significant performance improvements, its internal structure and computational mechanisms still exhibit certain limitations, necessitating further adjustments and improvements when applied in specific domains such as prediction of building energy consumption and renewable energy generation [25], [26], [27], [28]. The mask mechanism in the original Transformer design was primarily intended for accommodating the requirements of machine translation tasks. However, its applicability in power prediction tasks needs further exploration due to the presence of known auxiliary feature variables for future time steps. Additionally, the original attention computation mechanism of the Transformer consumes considerable computational resources and may lead to the model overlooking critical temporal information and local trends in the data [17], [18]. These factors significantly limit the model's ability to capture short-term dependencies in time series data. Moreover, the Self-Attention mechanism of the Transformer is unable to effectively learn fine-grained relationships between different auxiliary variables during computation, thereby weakening the model's ability to infer variable correlations when handling multivariate time series prediction tasks [29], [30].

Addressing the constraints of current research, this paper proposed a novel model named EMSformer, designed for multi-step short-term power prediction tasks in new power systems, aims at forecasting building loads demand and renewable energy. This is achieved through improvements to the Decoder structure and attention mechanism of the Transformer model. The main contributions of the model include:

- For scenarios in which auxiliary variable information is known in target variable prediction, the Decoder is innovatively improved, the attention mask mechanism is cancelled, and the related process is redesigned, so as to optimize the model's ability to capture the medium and long-term dependence and time correlation of time series.
- The simplification of structural design and the use of sparse attention mechanisms significantly reduce the computational complexity of the model. By performing Patch block processing on time series data before the multi-head sparse attention mechanism, and constructing a unique Patch Embedding structure, a "segment-to-segment" attention computing mode is realized. This not only improves the efficiency of the model in processing local information of time series but also considers the interaction between different variables. Moreover, in resource-constrained environments, the spatial and temporal complexity can be further reduced by adjusting the Patch sliding step size and sparsity factor, allowing the model to process longer-term time series data.
- By integrating innovative Patch Embedding technology and meticulously alternating the designed Self-Attention and Cross-Attention, the model's ability to learn local data features and reason about global trend changes is enhanced, significantly improving computational efficiency and prediction accuracy.
- A layer of BiLSTM was added to the back end of the Decoder of EMSformer to further capture the dynamic changes and dependencies of feature quantities in the time

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dimension. At the same time, this link also played a role in error correction, and further improved the stability and reliability of prediction by integrating multiple model outputs.

The remainder of the paper is organized as follows, Section II presents the EMSformer model and relevant theory; Section III covers the selection of datasets, the choice of evaluation metrics, the design of hyperparameters for the model, and other preparatory tasks before the experiment; Section IV compares EMSformer with other models and conducts a detailed correlation analysis to validate the effectiveness of the improvement strategies; Section V summarizes the paper and provides a perspective on future work.

II. The EMSformer Model Architecture

A. Problem Formulation

Utilizing historical power data, meteorological data, and some known forecasting information from the prediction period, the objective is to predict the power curve for a future time interval. Suppose the sample sequence of the electric power load (or renewable energy) in a certain region's power system is $\{t(1), t(2), t(3), \ldots, t(\mathrm{n})\}$. The information from these sampling points forms a set of feature data $\mathbf{D} = \left\{\left(\mathbf{X}^{(t-m)}, \mathbf{Y}^{(t-m)}\right), \left(\mathbf{X}^{(t-m+1)}, \mathbf{Y}^{(t-m+1)}\right), \ldots, \left(\mathbf{X}^{(t-1)}, \mathbf{Y}^{(t-1)}\right), \left(\mathbf{X}^{(t)}, \mathrm{Y}^{(t)}\right), \mathbf{Z}^{(t+1)}, \mathbf{Z}^{(t+2)}, \ldots, \mathbf{Z}^{(t+n)}\right\}$. In this context, t represents the current time point, where $\mathbf{X} = [\mathbf{X}^{(t-m)}, \mathbf{X}^{(t-m+1)}, \ldots, \mathbf{X}^{(t-1)}, \mathbf{X}^{(t)}] \in \mathbb{R}^{m\times d}$contains the historical data of relevant influence factors at time t, spanning m historical time points. d is the dimension of influencing factors, representing auxiliary variables related to renewable energy or load power. For example, solar power is usually affected by temperature, weather conditions, illumination intensity and other factors, while load power curve can indirectly reflect load demand through environmental factors, energy price information (day-ahead electricity price, actual electricity price, etc.) and day-ahead market demand information (preliminary load power reference value obtained from market settlement information). These factors have complex relationships with the forecasting target. By considering these auxiliary characteristics, the model can extract more relevant information. $\mathbf{Y} = [\mathbf{Y}^{(t-m)}, \mathbf{Y}^{(t-m+1)}, \ldots, \mathbf{Y}^{(t-1)}, \mathbf{Y}^{(t)}] \in \mathbb{R}^{m\times 1}$ corresponds to the actual power at the respective historical sampling points. $\mathbf{Z} = [\mathbf{Z}^{(t+1)}, \mathbf{Z}^{(t+2)}, \ldots, \mathbf{Z}^{(t+n)}] \in \mathbb{R}^{n\times k}$ represents the auxiliary variable information for n future time points at the current time t (such as the weather forecast data for the upcoming hours), where k is the known dimension of the influencing factors. $k \in d$. The predictive task at hand involves constructing a deep learning model that takes $\mathbf{D}$ as input, extracts key information, and predicts the power curve for the next n periods of time. This can be expressed as

$$Y^{\mathrm{t+1}} \sim Y^{\mathrm{t+n}} = f(\mathbf{D}) \tag{1}$$

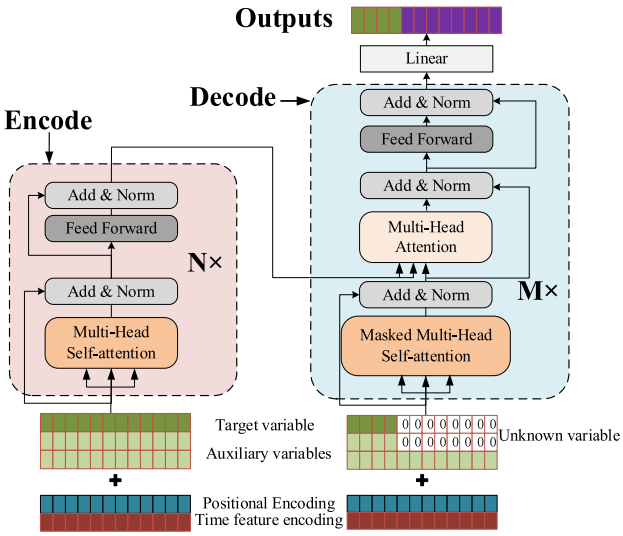


Fig. 1. The overall framework of transformer.

B. The Transformer Model and Sparse Attention Mechanism.

Compared to classical RNN, the Transformer model achieves highly parallelized sequence data processing through its unique self-attention mechanism, significantly speeding up the training process. This mechanism allows the model to dynamically capture complex dependencies between inputs and outputs. With its outstanding performance in handling time-series data, the Transformer model has also been successfully applied in the field of time series forecasting, demonstrating remarkable performance. The traditional model structure of the Transformer is illustrated in Fig. 1 [15].

Since the Transformer employs parallel computing and lacks positional information, it is necessary to add positional encoding and temporal encoding at the input end. This article uses a sinusoidal function for positional encoding to allow the model to better learn the positional information of the data, as shown in (2) to (3).

$$P_E(p, 2i) = \sin\left(p/(2L)^{2i/\mathrm{d}}\right) \tag{2}$$

$$P_E(p, 2i+1) = \cos\left(p/(2L)^{2i/d}\right) \tag{3}$$

The above formulas are the calculation methods for even and odd positions, respectively, where L is the length of the input sequence and d is the dimension of the features to be mapped.

The formula above can represent the relative positional relationships between data. Additionally, to express the absolute position of data in the time series, this article first decomposes the temporal information into year, month, day, hour, and minute features, then maps them into d-dimensional features through a fully connected layer. Finally, variable features are mapped into multi-dimensional features through one-dimensional convolution, and the three are summed together to form the input of the model.

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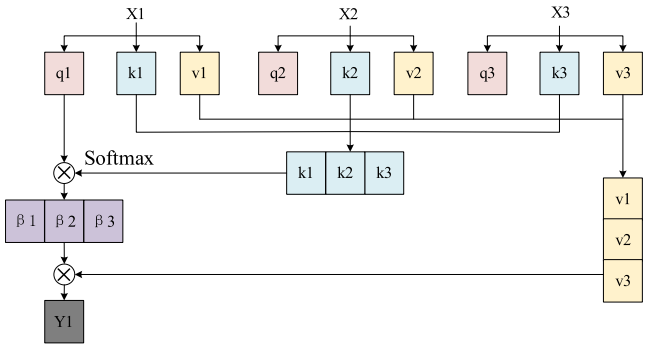


Fig. 2. Partial Self-Attention computation flowchart.

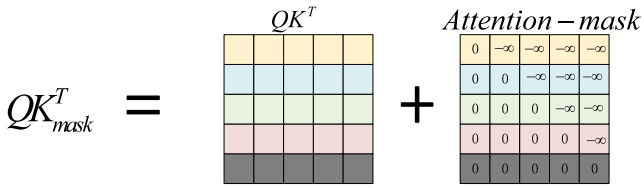


Fig. 3. The original intention of the mask mechanism is to prevent the leakage of future information. Decoder masking blocks future data access during sequence prediction.

The structure of the Transformer model is primarily composed of stacked combinations of Encoders and Decoders, which calculate the mutual influence between different variables through the Self-Attention mechanism, thus making it easier to solve long-range dependency issues. Suppose the input sequence to the Encoder is $\mathbf{X} = (x_1, x_2, \ldots, x_n)$, which is multiplied by three weight matrices to obtain the Query($\boldsymbol{Q}$), Key($\boldsymbol{K}$), and Value($\boldsymbol{V}$) vectors. The correlation between the input vectors is first calculated using the obtained $\boldsymbol{Q}$ and $\boldsymbol{K}$, usually in the form of a dot product. Then, an attention weight vector for all vectors on themselves is obtained, and a Softmax operation is performed on these weights so that the sum of the attention weights for all vectors on themselves is 1. Finally, a dot product with the $\boldsymbol{V}$ matrix yields the output. The calculation formula is shown as follows:

$$Attention(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = Softmax\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_K}}\right)\boldsymbol{V} \quad (4)$$

where d_K is the input dimension; $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{L \times d_K}$ represent Query, Key and Value in matrix form respectively. Multi-head attention mechanism is an advanced framework that integrates multiple Self-Attention modules, which can project input data into multiple independent subspaces for processing, each with different concerns. Through this multi-angle analysis, the model can simultaneously capture multiple semantic relationships of sequence data. The calculation process of partial Self-Attention is shown in Fig. 2. The probabilistic form of the attention coefficient for $q_i$can be expressed as:

$$A(q_i, \boldsymbol{K}, \boldsymbol{V}) = \sum_j \frac{k(q_i, k_j)}{\sum_j k(q_i, k_j)} v_j \quad (5)$$

$$k(q_i, k_j) = \exp\left(\frac{q_i k_j^T}{\sqrt{d_k}}\right) \quad (6)$$

The design goal of Transformer Decoder is to generate growth sequence prediction through a forward process, and use generative parallel computing prediction to solve the problem of high time complexity in long sequence data prediction. At the same time, in order to improve the reasoning speed, some historical sequences are added to guide the model reasoning. The Self-Attention layer of the Decoder is a masked Multi-head Self-Attention layer, and the masking principle is shown in Fig. 3. The original intention of the mask mechanism is to prevent the leakage of future information, ensure that the prediction model only makes decisions based on known historical data and current information, and ignore the influence of the fill part, so the weight after the current point in time is set to 0 in the attention calculation. However, in tasks such as power prediction, there is no problem of information leakage because future auxiliary variables are available. In fact, accurate future weather forecast data (NWP) can serve as a useful reference to guide the model's reasoning process for the present moment. For example, knowledge of future electricity prices may guide users to adjust their usage patterns. Therefore, in this situation, the traditional mask restriction may weaken the performance of the model. In view of this, the model proposed in this paper chooses to abandon the use of mask mechanism in order to make full use of all available information signals and enhance the accuracy and adaptability of the prediction.

Based on the above principles, the Transformer model can process data in parallel, greatly improving processing speed. However, the Self-Attention mechanism faces limitations when dealing with long sequences due to a time and space complexity of $o(L \wedge 2)$, which still poses high memory and time complexity constraints for the model when handling long sequence problems. Zhou et al. [17] found that the probability distribution in Self-Attention exhibits sparsity. and not all Queries provide great help to the model. Specifically, if the probability distribution of attention of a certain Query deviates from the uniform distribution to a large extent, Then this query can bring more valuable information and learning effect to the model. Conversely, if a query has a nearly uniform distribution of attention, its contribution to the model is relatively small, similar to directly taking the mean of Values as the output. Therefore, the *KL* divergence can be used to evaluate the 'sparsity' of the i-th query:

$$M(q_i, K) = \ln \sum_{j=1}^{L_K} \left[\exp\left(\frac{q_i k_j^T}{\sqrt{d_k}}\right)\right] - \frac{1}{L_K}\sum_{j=1}^{L_K} \frac{q_i k_j^T}{\sqrt{d_k}} \quad (7)$$

The first term in the equation is the log sum of evidence (LSE) for $q_i$with randomly sampled L_K keys, $L_K = c_1 \ln(L)$; The second term is its arithmetic mean. If the value is larger, it indicates that the attention weights corresponding to different keys have large differences, that is, the concentration degree of the probability distribution on different keys has significant differences, showing a high "diversity". This approach requires selecting the L_Q best Queries from the L available

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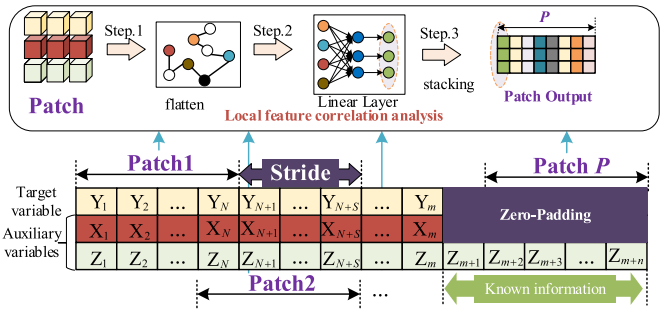


Fig. 4. Schematic of the Patch Embedding Operation.

to process with Key, $L_Q = c_2 \ln(L)$; while for the remaining Queries, use the average value of $\boldsymbol{V}$ as the output. Further simplification calculations can be made according to the relevant lemmas:

$$M(q_i, K) = \max\left(\frac{q_i k_j^T}{\sqrt{d_k}}\right) - \frac{1}{L_K}\sum_{\mathrm{j}=1}^{L_K}\frac{q_i k_j^T}{\sqrt{d_k}} \tag{8}$$

The probability sparse attention mechanism is an efficient method for extracting temporal information, which predicts which elements are important through a probability distribution, thus focusing only on these elements to reduce computational load. However, this mechanism also has some drawbacks: since the probability sparse sub-attention mechanism relies on the probability distribution to select important elements, inaccurate predictions of the probability distribution may lead to the loss of important information, thereby affecting the performance of the model; moreover, the Self-Attention matrix usually presents a long-tail distribution, meaning most values are very small with only a few values being larger, the probability sparse sub-attention mechanism needs to precisely identify these larger values, if not handled properly, it might overlook some key information.

C. Sparse Self-Attention Based on Patch Embedding

To enhance the Decoder's processing of local information, we introduce a Patch Embedding operation before each layer of attention calculation. The schematic is shown in Fig. 4. Specifically, assuming the original sequence has a length of L, it is divided into multiple small segments of length N, with each Patch containing the trend of changes in a segment of the sequence information. Next, all features within each Patch are flattened, and then these features are combined and learned through a linear layer to reduce the feature dimensionality while simultaneously capturing local information. This linear layer produces a new set of features by weighting and summing the input features and applying a bias term. This new set of features represents all the information within the Patch and. By learning among different features, enables the model to more easily grasp the interactions between different influencing factors as well as the local trends of change within a short period, thereby improving the prediction effect of the target variables. Ultimately, these transformed features are concatenated into a feature set representing the entire segment. This integrated set of features encapsulates the information and trends from the original sequence segments, so that the attention calculation is essentially performed on a "segment-to-segment" basis. This process strengthens the model's capability to process local information and is particularly suitable for scenarios involving multi-variable input and single-variable output tasks. In the field of energy forecasting, Patch Embedding also enables the processing of known variables during the prediction period, allowing the model to better learn the temporal trends and multivariate correlations of auxiliary variables.

Patch Embedding can be adjusted according to the actual forecast application. The size of the stride size S determines the number of patches and also affects the time and space complexity of subsequent attention calculation. The number of patches P is:

$$P = \frac{L - N}{S} + 1 \tag{9}$$

where P represents the number of patches of the processed time series, which constitute a new expression form after the segmentation and recombination of the time series data; S is the size of the sliding window or stride, that is, the jump interval between each patch, which is used to control how to sample the original signal to generate patches; N is the length of a single patch paragraph, that is, the number of time steps contained in each time series fragment. In general, a smaller (S) value can improve the performance of the model because it captures local features in the time series in more detail. However, with that comes higher computing costs.

In practice, to balance computational efficiency with performance, a probabilistic sparse attention mechanism can be combined with a patch-based processing framework. Through this mechanism, the model can focus on patches with more information, thereby reducing unnecessary calculations while maintaining performance. Under this configuration, even when the stride value. Under this configuration, even when the stride value1, meaning each patch is meaning each patch is closely connected, efficient computation can still be achieved. This approach prevents large strides between patches from causing discontinuities or gaps in the time series trends.

D. EMSformer Overall Framework of the Model

The overall structure of EMSformer model is shown in Fig. 5. The Decoder is used as the main structure in this paper, and the attention mechanism is innovated. The mask mechanism is no longer used in attention calculation, and the probabilistic sparse attention mechanism is adopted in both Self-Attention and Cross-Attention to select Q with strong information expression ability for attention calculation so as to reduce the amount of computation. The rest of Q is not calculated by dot product with V. For Q, which is not involved in the attention calculation, EMSformer takes two different approaches: 1) For Self-Attention, the model outputs the value vector (V) of the current moment directly, in order to preserve the original information; 2) For Cross-Attention, the model calculates the mean of the value vector (V) as the output, which can be seen as filling in the global information. The purpose of this is to interact with

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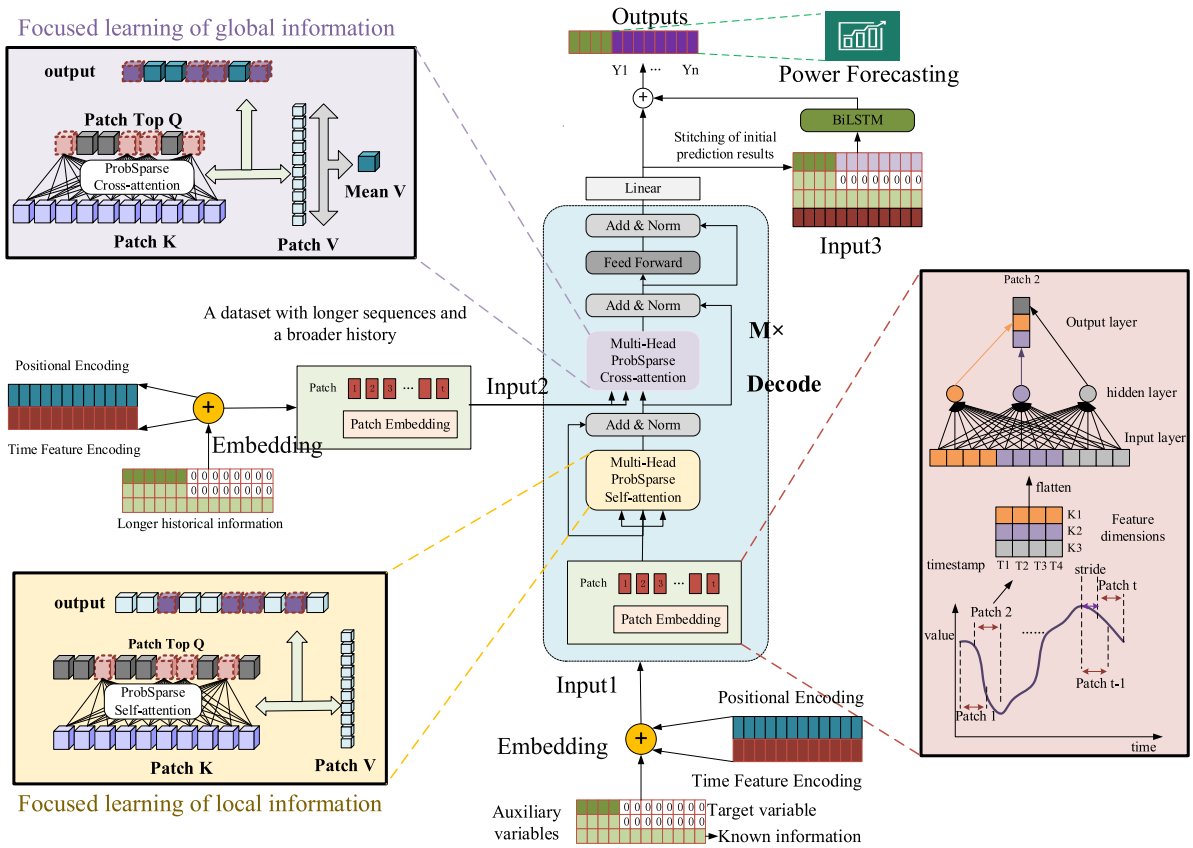


Fig. 5. Structure of EMSformer memory neutral network.

the Patch Embedding, where one focuses on dealing with local information and the other is responsible for learning the trend of global information.

In the EMSformer model, we have incorporated Patch Embedding operations before the Self-Attention calculations in both the input sequence and each layer of the Encoder. These operations are designed to capture local trend information at various temporal scales, with different patch segment lengths N used in the two instances.The data points processed through Patch Embedding not only carry local feature information but also undergo preliminary trend changes and variable interaction analysis. This preprocessing step allows the model to deeply understand the information of the auxiliary sequence before performing Self-Attention calculations. As a result, the EMSformer model can overcome the limitations of traditional Transformer models in dealing with auxiliary sequence correlation reasoning, leading to improved performance in processing multivariate time series data.

Combined with the introduction of probabilistic sparse attention mechanism, the model effectively extracts key features containing local information trend changes through segment-to-segment attention computation, while suppressing the interference of noise data. This mechanism allows the model to focus on important information in a more efficient way, using a single Query (Q) to represent specific paragraph information, rather than spreading attention across multiple Qs. By optimizing the interaction between Queries (Q) and Keys (K), EMSformer can accurately model the attention probability distribution, reducing the risk of losing important information due to inaccurate probability prediction. Furthermore, by adjusting the patch length N and the probabilistic sparsity factor, the model not only improves computational efficiency and saves computational resources but also enhances the model's robustness and reasoning ability. The Encoder part of the EMSformer no longer performs redundant attention calculations, providing the Decoder with preserved original sequence information and known features of the predicted day. For unknown variables and variables to be predicted, the model performs 0-padding processing. Unlike the input of the Decoder, the input of Encoder Input 2 allows historical sequences to have a longer length to capture long-term temporal dependencies. By introducing a Patch Embedding layer and setting a different patch length from the Decoder, the Encoder can capture local information on different time scales. With the cooperation of Self-Attention and Cross-Attention, corresponding to the sparse Self-Attention calculation where only part of the information is activated for subsequent processing, while the part that does not participate in the calculation directly passes its value (V), which helps to preserve the local trend of changes; in Cross-Attention, the part that does not participate in the calculation outputs the mean V to maintain the coherence of global information to capture a wide range of information trends or global context. This alternating computation strategy and complementary information flow design enable EMSformer to effectively extract detailed and global temporal features from

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TABLE I
DATASET INFORMATION

Datasets	Data Source	Auxiliary Features	Selected data time period	Data Granularity	Forecast Length	Target variable
Load1	The load dataset of Maine in the New England region of the United States	Day-ahead electricity price information, Scheduled load power reference, temperature, humidity, as well as actual electricity price information	2022.01.01—2024.01.01	1h	24	Electric load
Load2	The load dataset of a certain region in Australia.	Dry bulb temperature, wet bulb temperature, dew point temperature, humidity, electricity price.	2006.01.01—2008.09.27	30min	48	Electric load
Load3	Load History Data Translation for Arizona State University Tempe Campus	The weather information, including wind speed, temperature, and solar irradiance.	2022.01.01-2022.03.31	1h	24	Electric load
Load4	Load History Data Translation for Arizona State University Tempe Campus	The weather information, including wind speed, temperature, and solar irradiance, historical electric load and renewable energy.	2022.01.01-2022.03.31	1h	24	Net load
PW1	The dataset from a wind farm operated by the French National Railway Company.	Wind speed, wind direction, forecasted wind speed and wind direction, wind turbine orientation, actual measured speed	2018.05.01—2019.01.16	1h	24	Wind power
PW2	The GEFCom2014 Global Energy Forecasting Competition.	The day-ahead forecasted wind speed and wind direction at heights of 10 meters and 100 meters	2012.01.01—2013.12.01	1h	24	Wind power
PV1	The GEFCom2014 Global Energy Forecasting Competition.	Temperature, humidity, total irradiance, and other 12 NWP (Numerical Weather Prediction) information.	2012.04.01—2014.07.01	1h	24	Solar power
PV2	The 9A solar farm in Alice Springs, Australia.	Temperature, humidity, rainfall, total irradiance, actual values of scattered irradiance.	2017.10.20—2018.05.08	5min	1	Solar power

complex data, significantly enhancing the model's performance in time series prediction tasks.

Due to the model's use of parallel computing in inference to enhance computational efficiency, it does not perform reasoning across the temporal dimension, which can result in an inability to fully capture certain trends in the data. To address this, the final output layer of the decoder in this paper concatenates the decoder's preliminary predictions with the original features and then adds a BiLSTM layer. This enhances the model's ability to extract temporal features and also serves to correct errors in the decoder, improving the model's performance. The input information for Input 3 is not embedded but instead retains the original information. This approach reduces the complexity of inference and allows the model to learn more information through different processing methods applied to the raw data.

III. PREPARATION FOR COMPARATIVE EXPERIMENTS

A. *Description of the Experimental Platform and Dataset.*

In the forecasting process, there is no need to use the flow, line loss, or network architecture information of the power grid system. It only requires predicting the power curve by using data sets such as temperature, humidity, electricity price, and load characteristics at each node. In essence, various practical power system prediction applications involve predictive analysis on different datasets. Therefore, this paper does not experiment on the distribution network test system but directly verifies it on authoritative public standard test datasets, focusing on the validity and universality of the model and algorithm, and facilitating readers to conduct repeatability verification.

To validate the superiority and generalization performance of this model in power forecasting within novel electric power systems, a selection of datasets encompassing load energy consumption, photovoltaic power generation, and wind power generation were utilized for comparative experiments. These datasets not only chronicle energy consumption data (or renewable energy output data) but also furnish a wealth of auxiliary information such as weather data and electricity pricing information. Although the auxiliary variables provided are not consistent, they all can aid in model inference, allowing for a thorough validation of the superiority of the proposed model in reasoning with multiple feature variables. Secondly, the datasets used are all publicly available and have been studied by relevant scholars, ensuring the reliability of the data [31], [32], [33], [34], [35]. For detailed introductions to each dataset, please refer to Table I.

By collecting various data sources including weather forecasts, we have identified key factors affecting the prediction day, such as humidity, temperature, and electricity prices, which in turn allow for accurate prediction of the power curve for that day. When dealing with auxiliary sequences for Load1 and PW1, since they contain actual measured values and the actual data for the prediction period is unknown, we set their input values to zero to ensure that the model only makes predictions based on forecast data and historical real data, thereby simulating a realistic prediction environment. In Load3 and Load4, we test our model's performance on electric load and net load (total load minus renewable energy generation) forecasting tasks. When predicting net load, historical power data from electric load and new energy output are used as auxiliary variables with the prediction period set to zero; for PV2, its auxiliary variables are entirely composed of actual measurements, which are used to assess the model's performance in ultra-short-term predictions without any predictive information.

Before the experiment, datasets with missing values were filled, and datasets with excessive missing values were removed entirely for the entire day. The dataset was split into training, validation, and test sets at a ratio of 8:1:1. The training set was used for model parameter optimization, with Mean Squared Error (MSE) used as the loss function to evaluate model performance.

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The early stopping strategy was employed to update model parameters. To accelerate the convergence of weight parameters, enhance training efficiency, and improve generalization ability, the data was standardized, as shown in (10).

$$P_k' = \frac{P_k - \mu_k}{\sigma_k} \tag{10}$$

where P_k and P_k' are sample data of the k-type influencing factors before and after standardization. Through relevant calculations with the mean μ_k and standard deviation σ_k in historical data, the processed data can conform to the standard normal distribution. Compared with normalization, each sample point can affect the standardization, which can indirectly avoid the influence of outliers and extreme values through centering.

B. Comparison of Models and Hyperparameter Design

To validate the effectiveness of the proposed EMSformer model, this study selected six popular models for comparison: ARIMAX, XGBoost, LSTM, BiLSTM, Transformer, and Informer. The traditional machine learning models, ARIMAX and XGBoost, were tuned using grid search. For the Transformer, hyperparameters such as the number of layers in the Encoder and Decoder, model dimensions, and multi head attention were also tuned using grid search. To compare the generalization performance of the models, the hyperparameters of the remaining deep learning models were kept consistent across all datasets. All models (including EMSformer) had their hyperparameters set to be the same as those of the Transformer to ensure a fair comparison. The number of layers in LSTM and BiLSTM was set to be the same as the number of Decoder layers. The hyperparameters specific to EMSformer, such as the probability sparsity factor "Factor," were designed as per the original Informer paper. The number of layers in BiLSEM was also set to be the same as the number of Decoder layers. The hyperparameters for the patches in EMSformer are initially set based on empirical values derived from the periodic characteristics of the dataset; the length $N_1 = 4$ and stride $S_1 = 1$ for Input1-Patch1, and length $N_2 = 8$ and stride $S_2 = 2$ for Input2-Patch2, followed by detailed comparative analysis.

C. Evaluation Metrics

To assess the performance of various methods for point prediction results, this paper selects three widely-used deterministic prediction evaluation metrics: MAE, RMSE, and RSE:

$$\text{MAE} = \frac{1}{m}\sum_{t=1}^{m}|\hat{y}_t - y_t| \tag{11}$$

$$\text{RMSE} = \sqrt{\frac{1}{m}\sum\nolimits_{t=1}^{m}(\hat{y}_t - y_t)^2} \tag{12}$$

$$\text{RSE} = \frac{\sum_{t=1}^{m}(\hat{y}_t - y_t)^2}{\sum_{t=1}^{m}(\bar{y} - y_t)^2} \tag{13}$$

where m is the number of forecasting sample points; $\hat{y}_t$ is the predicted value at time t; y_t is the actual sample observation for the period t; $\bar{y}$ is the average value of the sample true values. This paper uses these three indicators to judge the accuracy of each model's day-ahead prediction error. The test set includes multiple day-ahead prediction results, and the mean of their error indicators is taken as the comprehensive evaluation metric for the model.

IV. Experimental Comparison Analysis

A. Model Comparison Analysis

In the day-ahead power prediction experiments conducted on multiple datasets, the Decoder inputs of each model included data from 72 historical time points as well as the corresponding known information for multi-step predictions. The input sequence for the Encoder contained 96 historical time series pieces of information (on top of which the EMSformer model also integrated auxiliary information for the multi-step predicted time points). For details on the predictive performance of each model on these datasets, please refer to Table II:

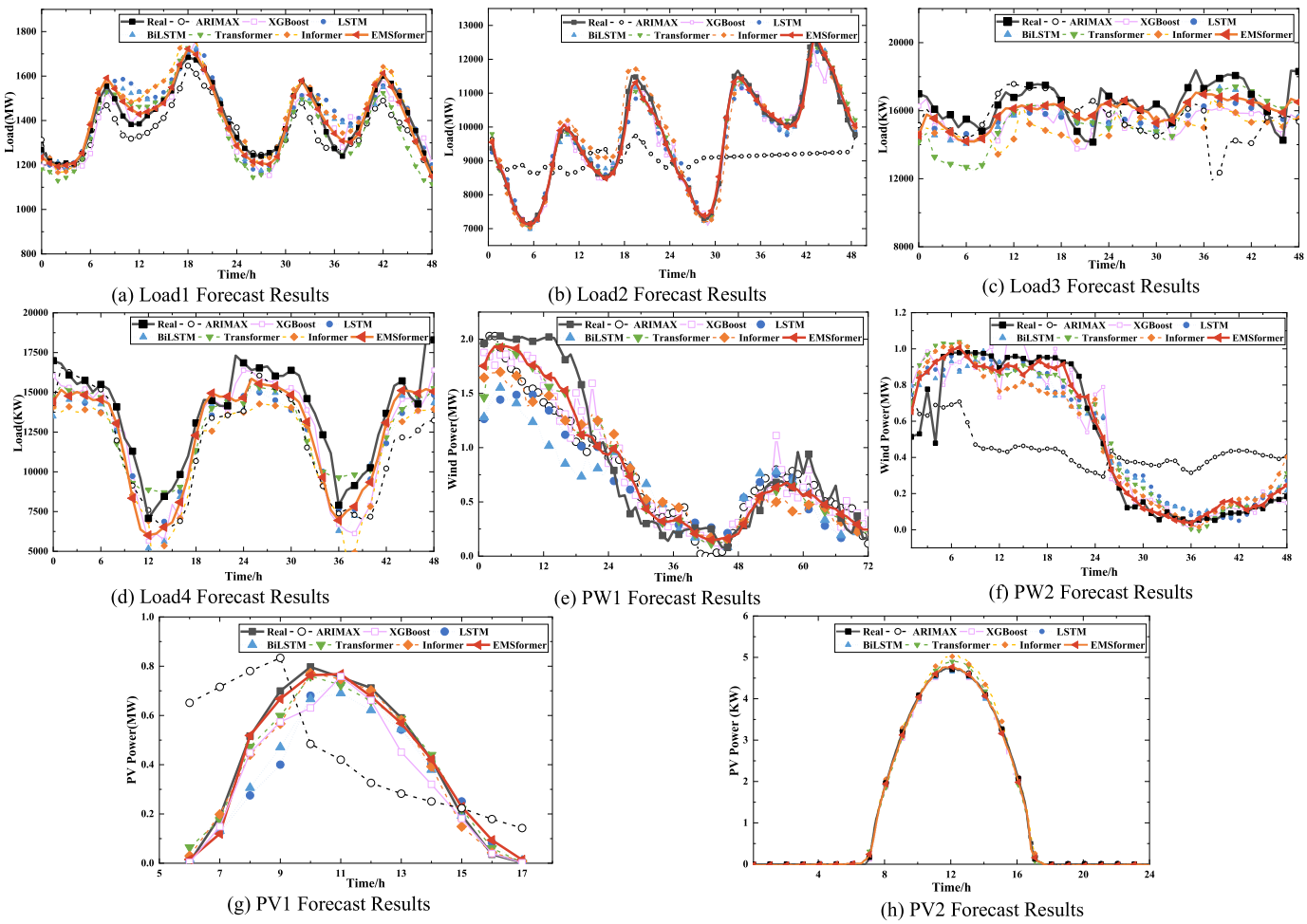
From the perspective of error results, EMSformer significantly improves prediction accuracy through enhancements to the Transformer architecture and achieves commendable performance across various datasets. As can be seen from the table, EMSformer not only achieves outstanding performance in day-ahead short-term predictions but is also applicable to the PV2 dataset. This result fully demonstrates that the performance improvement of EMSformer is not only due to the processing of Numerical Weather Prediction (NWP) data but also lies in its architecture's inherent high adaptability for power prediction tasks, allowing the model to more easily excavate the temporal characteristics between different related factors. Moreover, compared to Transformer and Informer, both memory consumption and computational time complexity are significantly reduced. A detailed analysis of the prediction results of different models shows that some predictions and the trend of MAE variation are illustrated in Figs. 6 and 7(selected time periods: Load1: 2023/11/29∼2023/11/30; Load2: 2008/08/03∼2008/08/04; Load3 and Load4: 2022/3/24∼2022/3/25; PV1: 2014/4/30; PV2: 2014/4/30; PW1: 2018/12/01∼2018/12/03; PW2: 2013/10/02∼2013/10/03). The ARIMAX model generally performed the worst in prediction tasks, primarily due to error accumulation during multi-step forecasting and inflexible handling of external variables, leading to inaccurate results. Other models showed varying strengths and weaknesses across different datasets. For example, the Transformer model performed well on complex datasets, while LSTM and BiLSTM did better on datasets with clear temporal trends. XGBoost was more efficient on datasets with strong linear relationships. The EMSformer model proposed in this paper has achieved good prediction results across all tested datasets, demonstrating its superiority and generalization capabilities in the field of energy forecasting. A detailed analysis will follow.

Fig. 6(a) and (b) clearly demonstrates the outstanding ability of the EMSformer model in capturing the dynamic changes of load power. The predicted results of the model highly align with the actual load power curve, especially excelling in peak and valley predictions compared to other models. By deeply

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TABLE II
ERROR EVALUATION RESULTS UNDER LOAD FORECASTING AND RENEWABLE ENERGY OUTPUT PREDICTION TASKS

Dataset	Load Dataset											
	Load1			Load2			Load3			Load4		
Metrics	MAE	RMSE	RSE	MAE	RMSE	RSE	MAE	RMSE	RSE	MAE	RMSE	RSE
ARIMAX	90.499	117.790	0.385	1204	2214	2.523	1372	1766	1.617	1601	2060	0.517
XGBoost	58.597	85.290	0.202	208	289	0.043	1267	1501	1.167	1419	1778	0.385
LSTM	64.111	84.860	0.233	237	317	0.052	1238	1450	1.089	1792	2185	0.582
BiLSTM	64.904	98.055	0.267	201	267	0.036	1122	1319	0.901	1601	1946	0.462
Transformer	59.823	89.652	0.223	202	274	0.038	1335	1531	1.215	1656	1954	0.465
Informer	60.968	89.747	0.223	228	304	0.047	1709	1990	2.051	2473	2918	1.039
EMSformer	**49.831**	**78.721**	**0.172**	**163**	**217**	**0.024**	**1091**	**1311**	**0.890**	**1396**	**1657**	**0.335**
Dataset	Renewable Energy Dataset											
	PW1			PW2			PV1			PV2		
Metrics	MAE	RMSE	RSE	MAE	RMSE	RSE	MAE	RMSE	RSE	MAE	RMSE	RSE
ARIMAX	0.149	0.213	0.305	0.206	0.261	0.779	0.152	0.365	2.804	0.068	0.216	0.015
XGBoost	0.115	0.172	0.199	0.143	0.193	0.427	**0.032**	0.076	0.124	0.114	0.279	0.025
LSTM	0.134	0.195	0.257	0.140	0.190	0.414	0.045	0.078	0.128	0.064	0.180	0.010
BiLSTM	0.149	0.219	0.323	0.138	0.186	0.396	0.043	0.076	0.122	0.067	0.182	0.011
Transformer	0.110	0.159	0.171	0.137	0.184	0.389	0.044	0.085	0.154	0.092	0.199	0.013
Informer	0.134	0.185	0.231	0.158	0.211	0.511	0.040	0.084	0.149	0.129	0.289	0.027
EMSformer	**0.109**	**0.154**	**0.161**	**0.120**	**0.166**	**0.316**	0.034	**0.070**	**0.104**	**0.064**	**0.172**	**0.009**



(a) Load1 Forecast Results (b) Load2 Forecast Results (c) Load3 Forecast Results

(d) Load4 Forecast Results (e) PW1 Forecast Results (f) PW2 Forecast Results

(g) PV1 Forecast Results (h) PV2 Forecast Results

Fig. 6. Comparison of forecasting results by different models across multiple datasets.

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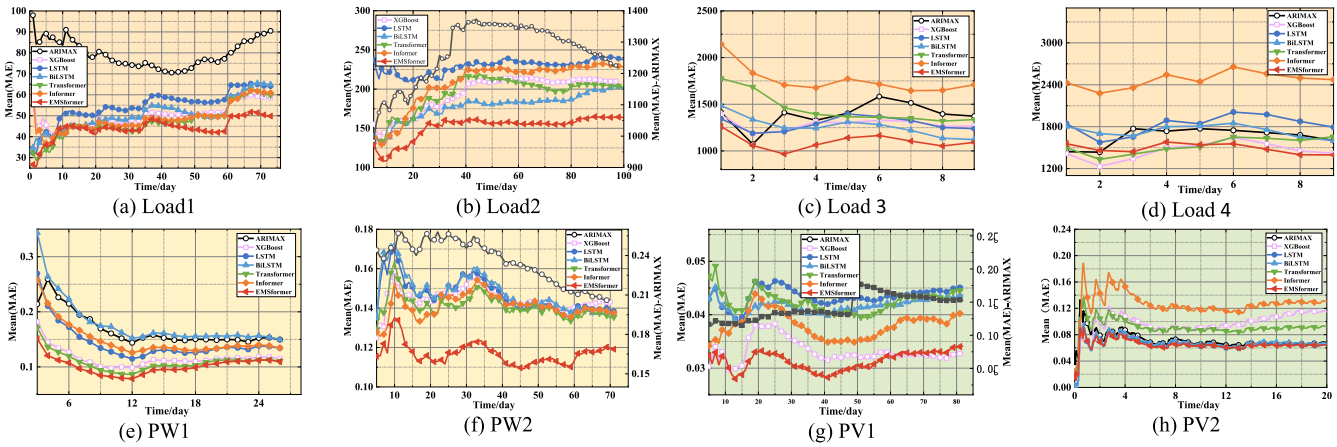


Fig. 7. Comparison of average MAE error trends across multiple datasets for different models.

analyzing and intelligently processing historical data along with information such as electricity prices, temperature, and humidity during the forecast period, the model effectively extracts key features from the time series and accurately predicts future load trends. Observing Fig. 7(a), it is evident that as time progresses, changes in statistical properties of the data make it difficult for traditional models to adapt, whereas the EMSformer, with its integrated BiLSTM layer, can better capture the seasonal impact on the load and shows reduced MAE during specific periods (such as days 40 to 60), confirming its advantage in generalization performance and compensating for the lack of temporal reasoning in the Transformer. Moreover, the comparative analysis in Fig. 7(b) over the first 30 days also highlights EMSformer's significant superiority in short-term prediction accuracy. compared to the Transformer, it reduces RSE by 22.8% and 36.8% on the Load1 dataset and Load2 dataset, reflecting its enhanced capability to excavate temporal characteristics.

Fig. 6(c) and (d) shows the prediction results of the model for electrical load and net load power during the same period. Due to the large fluctuations in the electrical load dataset, it is difficult to capture potential uncertainties from weather data alone. The performance of EMSformer at certain moments is not very well fitted, but overall, EMSformer basically fits the trend of load changes, and its overall error still maintains the best performance among comparative models. Compared with Transformer and Informer, the improvement effect is significant. On Load3 and Load4 datasets, EMSformer reduced the RSE by 26.7% and 27.9% respectively compared with Transformer, further verifying the effectiveness of the improvements made by EMSformer.

From Fig. 6(e) and (f), it can be seen that in the 24-hour wind power prediction task, due to the randomness of wind speed and direction, there is considerable deviation between the forecasted and actual values of wind speed and direction in weather forecasts, making it difficult to predict the fluctuation of wind power. Traditional models such as BiLSTM and LSTM have failed to effectively capture the changing trends of wind power. In contrast, EMSformer, by integrating Patch Embedding technology, effectively captures both the local details and global trends of wind speed and direction. Moreover, it avoids the traditional mask mechanism, thus providing a more comprehensive analysis of future wind conditions and offering precise data support for prediction. Although Fig. 7(f) demonstrates the excellent performance of EMSformer on the PW2 dataset, Fig. 7(e) indicates that due to the limitations of weather forecast data, the difference in MAE between EMSformer and Transformer is not significant. This reflects that data quality is a key bottleneck for improving prediction performance for all models. However, even under limited data quality conditions, the innovative approach of EMSformer still helps to improve prediction accuracy. Compared to the traditional Transformer structure, EMSformer simplifies the model architecture by abandoning typical Encoder configurations, thereby reducing model complexity. At the same time, by adopting sparse attention mechanisms, it effectively extracts critical feature information while maintaining low computational resource consumption. On the PW1 and PW2 datasets, EMSformer reduces RSE by 5.8% and 18.7% respectively compared to Transformer, demonstrating its efficiency and significant potential in wind power prediction while reducing resource requirements.

In the PV1 dataset, the errors observed in Fig. 7(g) and Table II indicate that in some cases, the Informer outperforms the Transformer. Leveraging its probabilistic sparse attention mechanism, Informer not only reduces computational complexity but also maintains efficient capturing of crucial time-point information. Building upon this, EMSformer further optimizes the encoding process of time series data by transforming input data through Patch Embedding technology, thereby enhancing the model's understanding and representation of semantics within time segments. Moreover, EMSformer incorporates the temporal reasoning advantages of BiLSTM, effectively utilizing sequential information in scenarios with stable photovoltaic output trends or ultra-short-term forecasting. Results show that on the PV1 and PV2 datasets, EMSformer reduced the RSE by 32.4% and 30.7% compared to the Transformer, confirming its significant advantage and accuracy in photovoltaic power prediction.

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TABLE III
THE IMPACT OF THE MASKING MECHANISM ON THE PREDICTIVE PERFORMANCE OF THE MODEL

Model	Removing Mask and Outputting V in Informer Decoding		Removing Mask in Informer Decoding		Informer		Removing Mask in Transformer Decoding		Transformer	
Metric	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Load1	**56.443**	**85.062**	57.638	87.367	60.968	89.747	**57.831**	**89.429**	59.823	89.652
Load2	**182**	**241**	198	258	228	304	**193**	**256**	202	274
Load3	**1352**	**1585**	1382	1608	1709	1990	1523	1721	**1335**	**1531**
Load4	**1875**	**2259**	1955	2300	2473	2918	**1630**	**1924**	1656	1954
PW1	**0.114**	**0.160**	0.114	0.165	0.135	0.185	**0.110**	**0.158**	0.110	0.159
PW2	**0.134**	**0.181**	0.139	0.183	0.158	0.211	0.149	0.201	**0.137**	**0.184**
PV1	0.043	0.084	0.042	**0.081**	**0.040**	0.084	**0.044**	**0.083**	0.044	0.085
PV2	**0.091**	**0.206**	0.097	0.215	0.129	0.289	**0.091**	**0.199**	0.092	0.199

Experiments on multiple datasets demonstrate a significant improvement in prediction accuracy of EMSformer compared to other advanced models such as Transformer and Informer. Overall, EMSformer achieves deep analysis of time series data and efficient information extraction by integrating innovative Patch Embedding technology, finely designed Self-Attention and Cross-Attention mechanisms, BiLSTM residual correction techniques, and the idea of learning future information trends without masking. This showcases its outstanding capabilities in advanced time series analysis tasks.

B. Comparison of MASK Mechanism

To analyze the impact of removing the mask mechanism on the model, the errors of Transformer and Informer were compared before and after removing the mask, as shown in Table III. Overall, it can be seen that removing the mask is helpful for improving the predictive performance of the models, with particularly significant effects observed on Load2 and PW1 datasets. For instance, the original Informer had an MAE error of 228 and an RMSE of 304 on Load2, which decreased to an MAE of 198 and an RMSE of 258 after removing the mask, demonstrating a notable improvement over Transformer. This also indicates some limitations of the original Informer in using probabilistic sparse attention in the Decoder.

Informer adopts a probabilistic sparse pattern to select a subset of key-value pairs for attention computation. By introducing a probability distribution, the model can selectively attend to those keys deemed "important" rather than mechanically computing all keys. However, this strategy implicitly allows Queries (Q) to interact with subsequent Keys (K) and select Queries (Q) based on this probability distribution. Yet, under this mechanism, the use of a mask is still employed to prevent information leakage. As a result, the predicted probability distribution of Q may not align with the actual probability distribution that can participate in the computation. This discrepancy is particularly evident when there are known auxiliary variables in the prediction time period, leading to biases in estimating the probability distribution of Q. Removing the mask effectively addresses this issue, enabling the Informer to better learn the trends of future time periods, utilize global information to capture long-term dependencies in the sequence, and reduce excessive attention to irrelevant information.

When performing single-step ultra-short-term predictions with limited auxiliary variable information (or when the auxiliary variables in the prediction time segment are unknown), Cross-Attention computation with the Encoder allows Q to learn information from all K, which is why the accuracy improvement is not significant for Transformer and Informer on PV2.Overall, it can be concluded that removing the mask mechanism in Decoder attention computation is beneficial when auxiliary variables in the prediction time period are known.

Furthermore, in the probabilistic sparse attention calculation process of the Informer decoder, for the parts that do not participate in the calculation, the method proposed by the EMSformer in this paper is applied. That is, the V value at the current moment is directly outputted to fully preserve the original information, so as to verify the impact of this method on the prediction accuracy. It is found that there are significant improvements on most of the datasets, and this method even outperforms the Transformer on some datasets. This indicates that when the auxiliary variables in the prediction time period are known, the improved strategy is effective for the prediction tasks of load power and new energy generation power.

C. Analysis of Patch Embedding Hyperparameters

The analysis of Patch Embedding hyperparameters for EMSformer on the Load1 dataset focused on the effect of Patch lengths N_1 and N_2 on accuracy, keeping other parameters constant as in Section IV-A. MSE results under varying Patch lengths are shown in Fig. 8, with the baseline in a black box and optimal N_1/N_2 combinations in an orange box. Although the Section IV-A settings didn't maximize Patch Embedding's potential, they still outperformed other models, confirming its importance in feature extraction and EMSformer's superiority in precise time series prediction. The error distribution highlights that N_1 is crucial for local information analysis; its careful selection is essential to balance capturing detail without losing local context, affecting model inference at finer scales.

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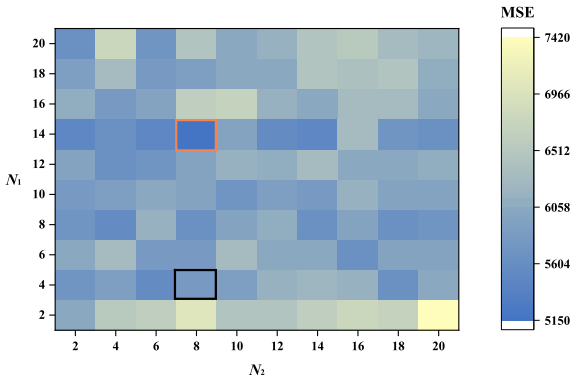


Fig. 8. The errors of the model under different N_1 and N_2 values.

D. Model Comparison for Probabilistic Forecasting Tasks

The EMSformer is not only applicable to point prediction tasks but also suitable for interval predictions. By adopting quantile regression, the loss function is replaced with a pinball loss function:

$$pinball = \begin{cases} (1-\tau)(\hat{y}_{t,\tau} - y_t), & \text{if } y_t < \hat{y}_{t,\tau} \\ \tau(y_t - \hat{y}_{t,\tau}), & \text{if } y_t \geq \hat{y}_{t,\tau} \end{cases} \tag{14}$$

where τ represents the quantile, $\hat{y}_{t,\tau}$ is the predicted value at quantile τ. The interval bounded by the predicted values at 0.05 and 0.95 corresponds to α 90% confidence interval for the forecast. The interval predictions are evaluated using the pinball loss, Predicted Interval Coverage Probability (PICP), and Prediction Interval Normalized Average Width (PINAW). PICP is used to assess the coverage of the actual values within the predicted intervals, while PINAW is used to evaluate the width of these intervals. The formulas for calculating PICP and PINAW are as follows:

$$\text{PICP} = \frac{1}{n}\sum_{t=1}^{n} c_t, c_t = \begin{cases} 1, & \text{if } y_t \in [L_{t,a}, U_{t,a}] \\ 0, & \text{if } y_t \notin [L_{t,a}, U_{t,a}] \end{cases} \tag{15}$$

$$\text{PINAW} = \frac{1}{nR}\sum_{t=1}^{n} (U_{t,\alpha} - L_{t,\alpha}) \tag{16}$$

where $U_{t,a}$ and $L_{t,a}$ represent the upper and lower bounds of the prediction interval with a confidence level of α. n denotes the number of forecast samples, while R is the range of variation in the predicted target values, used for normalizing the average bandwidth.

A comparison was made between one-hour-ahead probability forecasts and 24-hour probability forecasts on the load1 dataset. The forecast results are shown in Table IV. As can be seen from the table, EMSformer maintains high accuracy in probability prediction tasks. When the width of the prediction interval is very narrow, its coverage will decrease, leading to a conflict between the two. Therefore, there may be situations where individual PICP or pinaw values are not as good as other models. However, as indicated by the pinball measure, the superiority of EMSformer's prediction accuracy is evident. It can achieve both narrower intervals and higher coverage rates, proving the superiority of the proposed prediction method. In contrast, EMSformer had a superior prediction interval. This further demonstrates EMSformer's outstanding ability in time series analysis tasks.

TABLE IV
INTERVAL FORECAST RESULTS FOR LOAD 1

CI	$\alpha = 40\%$					
TASK	1-H-AHEAD TASK			24-H-AHEAD TASK		
	pinball	PICP	PINAW	pinball	PICP	PINAW
ARIMAX	21.95	0.394	0.017	78.45	0.285	0.059
XGBoost	26.77	0.382	0.018	51.20	0.264	0.027
LSTM	22.15	0.374	0.015	55.23	0.277	0.029
BiLSTM	20.38	0.374	**0.012**	52.49	0.285	0.024
Transformer	23.73	0.305	0.013	50.40	0.268	0.021
Informer	22.02	0.354	0.013	52.47	0.281	0.024
EMSformer	**17.24**	**0.513**	0.016	**45.52**	**0.289**	**0.019**
CI	$\alpha = 80\%$					
TASK	1-H-AHEAD TASK			24-H-AHEAD TASK		
	pinball	PICP	PINAW	pinball	PICP	PINAW
ARIMAX	11.39	0.752	0.034	39.66	0.650	0.092
XGBoost	13.86	0.648	0.040	38.66	0.470	0.052
LSTM	13.19	0.759	0.042	31.23	**0.679**	0.075
BiLSTM	12.48	0.690	0.034	34.15	0.608	0.067
Transformer	13.28	0.720	0.035	30.44	0.624	0.063
Informer	13.41	0.761	0.038	32.82	0.648	0.073
EMSformer	**9.59**	**0.768**	**0.032**	**30.23**	0.609	**0.049**
CI	$\alpha = 90\%$					
TASK	1-H-AHEAD TASK			24-H-AHEAD TASK		
	pinball	PICP	PINAW	pinball	PICP	PINAW
ARIMAX	6.940	0.881	0.047	43.13	0.706	0.119
XGBoost	9.608	0.836	0.049	29.89	0.610	0.076
LSTM	8.949	0.839	0.056	23.62	**0.813**	0.129
BiLSTM	8.911	0.806	0.050	24.13	0.742	0.099
Transformer	8.883	0.830	0.050	23.15	0.715	0.075
Informer	8.733	**0.883**	0.052	22.96	0.769	0.104
EMSformer	**6.438**	0.827	**0.037**	**22.06**	0.735	**0.069**

V. CONCLUSION

The multi-step prediction of building energy consumption and renewable energy generation plays a crucial role in power system, providing essential information support for grid planning and operation. This includes applications in distributed energy management, load-side demand response management, equipment optimization scheduling, power flow calculation, and more.

In this study, we proposed EMSformer, a novel short-term power multi-step prediction model based on the Transformer-BiLSTM-Patch architecture. The original Transformer model has been enhanced to better address prediction tasks within energy systems, thereby improving its performance in such tasks. The EMSformer model, with its unique architecture and meticulous design, significantly outperforms other benchmark models such as Transformer, Informer, and LSTM. Under the same set of hyperparameters, it achieves notable improvements in predictive performance across multiple datasets. This highlights its exceptional ability to capture complex relationships between variables, generalization capabilities, and prediction

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accuracy. Notably, the EMSformer excels in efficient utilization of computational resources, achieving lower computational resource consumption compared to models like Transformer and Informer. This enhances EMSformer's adaptability in resource-constrained environments and improves its deployment flexibility across different application scenarios. Additionally, The Patch Embedding and its improvement strategies proposed in this paper have been validated on multiple datasets, demonstrating good generalizability. This study provides valuable references for other model designs, and scholars can draw inspiration from its structure to optimize their models for handling other time series tasks. The hyperparameters of the Patch, such as lengths N1 and N2, can be adjusted using common machine learning methods such as swarm intelligence optimization algorithms and grid search, to further enhance the model's adaptability to different datasets and tasks.

In future work, the model can be improved in the following aspects. Although EMSformer outperforms Transformer under the same hyperparameter settings, to maximize its performance in practical applications, it still requires, like other predictive models, machine learning techniques for hyperparameter optimization (such as model dimensions, number of layers, and Patch lengths N_1 and N_2), which can be time-consuming. This is a common issue in the field of artificial intelligence. We plan to further explore adaptive hyperparameter adjustment mechanisms to allow the model to automatically tune according to dataset characteristics, making it widely applicable to various multi-feature time series prediction tasks. Additionally, this will provide innovative ideas for deep learning hyperparameter optimization. Furthermore, we will develop a federated learning-based multi-energy joint prediction model to address the needs of more complex energy internet systems.

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